

Stylus Common UI Migration Guide

Product Version 26.10

May 2026

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Printed in the United States of America.

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About This Manual

This document applies to and provides information about Stylus Common User Interface.

How This Document Is Organized

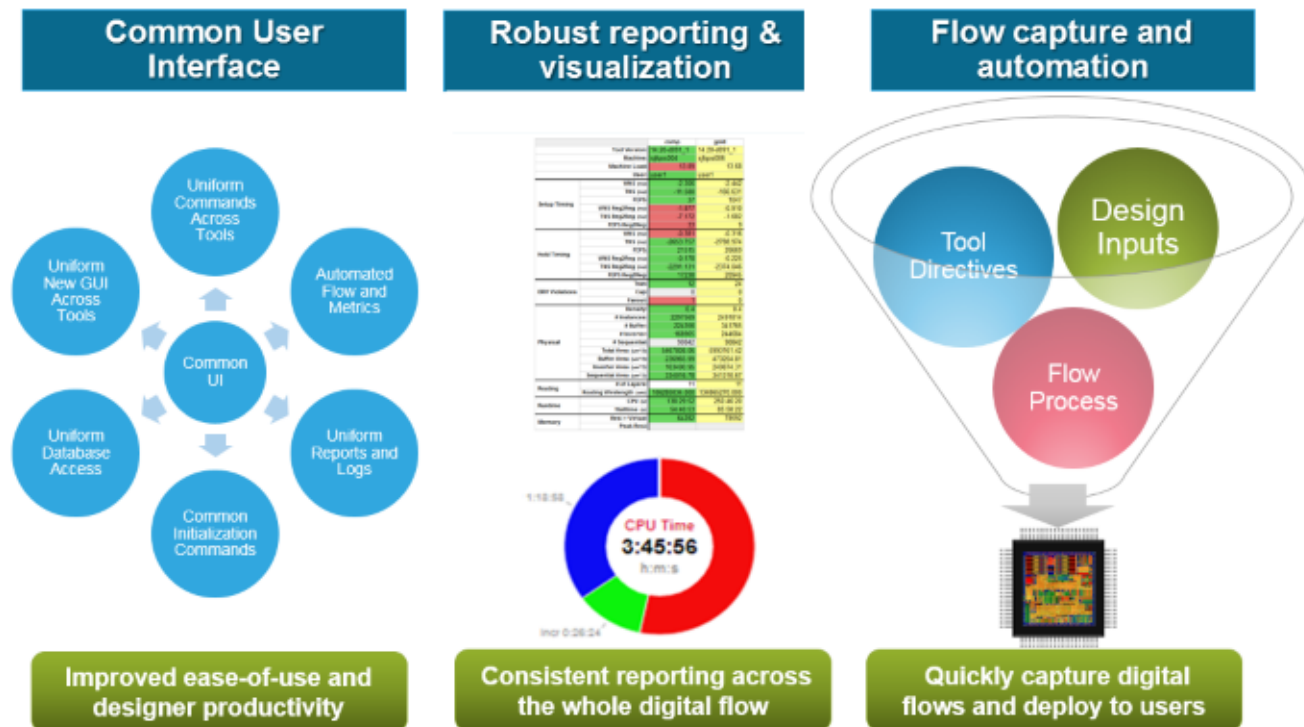
This manual is organized into chapters that cover broad areas of Stylus Common UI functionality. Each chapter contains topics that may address one or more of the following areas:

- Overview of the Stylus Common UI functionality
- Key areas of difference between legacy and Stylus Common UI, including design initialization, database access, flow, and timing reports.

Introduction to Stylus

Stylus is an infrastructure that offers three significant features:

- Stylus Common User Interface offering consistent commands across the whole digital flow
- Stylus Unified Metrics for capturing and reporting
- Stylus Flow Kit for design flow capture and deployment



The Stylus Common User Interface (Common UI from this point onwards) has been designed to be used across Genus, Joules, Modus, Innovus, Tempus, and Voltus tools. By providing a common interface from RTL to signoff, Common UI enhances user experience by making it easier to work with multiple Cadence products.

Common UI simplifies command naming and aligns common implementation methods across Cadence digital and signoff tools. For example, the processes of design initialization, database access, command consistency, and metric collection have all been streamlined and simplified. In addition, updated and shared methods have been added to run, define, and deploy reference flows. These updated interfaces and reference flows increase productivity by delivering a familiar interface

across core implementation and signoff products. You can take advantage of consistently robust RTL-to-signoff reporting and management, as well as a customizable environment.

The key features of Common UI are:

- **Common database access methods in all tools**

You can now use a single command, `get_db`, to query and filter all database attributes. `get_db` replaces `dbGet`, `get_ccopt_property`, and various other `get` methods. It includes `get_property` timing attributes, although `get_property` is still retained for SDC usage.

The `set_db` and `reset_db` commands are the companion commands to set values.

For more information, refer to the [Working with `get\_db` and `set\_db`](#) chapter of the *Stylus Common UI Migration Guide*.

- **Common initialization flow with common MMMC file**

The initialization flow is now the same for all the tools and uses the same MMMC file.

A new `timing_condition` object has been added to the existing MMMC syntax in the MMMC file. This object is required by Common UI and makes it possible to:

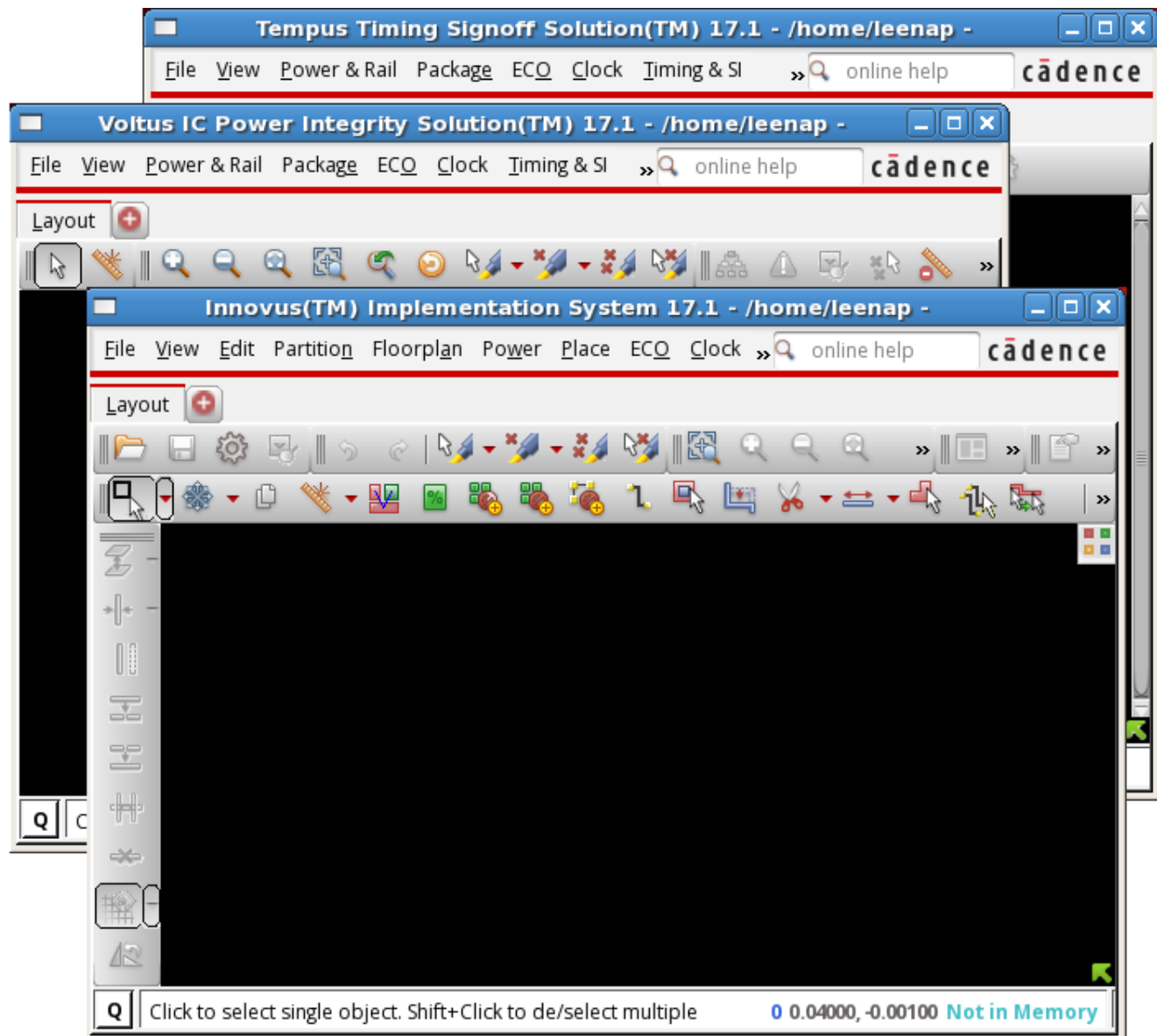
- Remove all timing data from `power_intent` files
- Bind `power_domains` to the MMMC timing data more easily.

For more information, refer to [Stylus Common UI Initialization](#) section and the following initialization-related commands in the *Text Command Reference*:

- `read_mmmc`
- `read_physical`
- `read_netlist`
- `read_power_intent`
- `init_design`

- **Common GUI**

Common UI provides a uniform graphical user interface across tools with consistent menus, widgets, and forms. The menus are organized in the same order, although each tool only shows the menus or sub-menus for commands supported by that tool. The widgets and forms for common commands and windows are the same.



- **Common timing reports**

In Common UI, the `report_timing` command has been enhanced to:

- Allow fast identification of issues related to clock definition, optimization, and constraints.
- Facilitate analysis and debugging. Enhancements include easier issue identification, cut and paste option, similar reports for different tools, and some customization features.
- Produce a more efficient and consistent report format. This includes aligning similar data from launch and capture paths and aggregating useful information that may not be visible in detailed paths.

For more information, refer to the [Timing Reports](#) section.

- **New flow kit**

A new set of flow commands, including `create_flow` and `create_flow_step`, support user flows more easily. New flow and flow_step objects store the state of the flow with the database.

For more information, refer to the [Foundation Flows](#) section.

- **Unified metrics**

Common UI enables you to generate unified and holistic metrics, which make it easier to summarize and compare results and lead to effective debugging. For information on new metric commands and attributes, refer to the [Metrics](#) section.

- **Consistent command naming across tools**

Common UI uses consistent command naming conventions across tools.

- All commands and options are now in lower case with an underscore as a separator.

You can find the Common UI equivalent of any legacy command by using the `get_common_ui_map` command. For example, you can find the Common UI equivalent of the legacy `read_lib` command as follows:

```
tempus 5> get_common_ui_map read_lib
Scheduling timing library file(s) '-help' to be loaded when the init_design
command is issued
read_lib read_libs
-aocv                -aocv_libs
-cdb                  -noise_libs
-lef                  -lef_files
-max                  -max_libs
-max_cdb              -max_noise_libs
-min                  -min_libs
-min_cdb              -min_noise_libs
-oaRef                -oa_ref_libs
-pgv                  -power_ground_files
-socv                 -socv_libs
libFiles              in_file
```

You can use the `eval_legacy` command to run a legacy command in the legacy Tcl interpreter. In normal usage, this command is not required. However, it may be needed in cases where some legacy usage is not yet available in Common UI. This command is intended as a temporary workaround as users transition to Common UI and will be removed in a future release.

- **Consistent usage across tools**

- All global values are now root attributes and accessed the same way across tools.

The previous `set_xxx_mode` options and global Tcl vars are replaced with root attributes that are accessed/modified with `get_db/set_db/reset_db`. You can check the mapping of these legacy mode commands by using the `get_common_ui_map` command. For example, you can check the mapping for the legacy `set_design_mode` command as follows:

```
tempus 2> get_common_ui_map set_design_mode

set_design_mode                set_db

-addPhysicalCell               add_filler_cell_name_style

-expressRoute                  design_express_route

-flowEffort                    design_flow_effort

...
```

For example, the legacy command

```
tempus 1> set_design_mode -flowEffort high
```

is

```
tempus 1> set_db design_flow_effort high
```

in Common UI.

You can also check the mapping for any legacy Tcl global in the same way:

```
tempus 2> get_common_ui_map dbgLefDefOutVersion

dbgLefDefOutVersion            write_def_lef_out_version
```

For example, the legacy command

```
tempus 1> set dbgLefDefOutVersion 5.7
```

is

```
tempus 1> set_db write_def_lef_out_version 5.7
```

in Common UI.

You can use the `convert_legacy_to_common_ui` command to convert old legacy Tcl scripts into Common UI Tcl scripts. Although this command does not handle 100% of the legacy commands, it does handle most of the legacy commands and adds comments where manual checking or fixing is needed.

- A single `help` command can now be used to obtain information about attributes.

commands, messages, and objects. The `help` command facilitates discovery and self-help. It supports pattern matching across command, object and attribute names.

For more information on using , see [help](#).

o

- **Uniform logging**

Command logging plays a vital role in the debug process. Stylus Common UI provides uniform logging across products by logging all commands in the log file, irrespective of whether they are issued interactively or through startup files and scripts.

For more information, refer to the [Uniform Logging](#) section.

Related Information

- [Stylus Common UI Initialization](#)
- [Database Access and Handling](#)
- [Foundation Flows](#)
- [Timing Reports](#)
- [Uniform Logging](#)
- [Uniform Startup](#)
- [Metrics](#)

Stylus Common UI Initialization

Initialization is the first step in Stylus Common UI. It is the process of reading libraries, physical data, HDL/netlist, power intent, and constraints. After initialization, the design is in a consistent state, and is ready for manipulation or timing.

Earlier, various tools used different initialization strategies. For example, Genus used an incremental approach, with each command being an autonomous operation. Once the libraries were loaded, the user could query the libraries. On the other hand, Innovus uses a setup or load strategy where variables are first defined pointing to the various design elements. Then, an initialization command is issued and all the elements are loaded at once. Tempus supports versions of both these strategies.

In Common UI, there is a single initialization methodology that works across Genus, Innovus, and Tempus. Common UI provides two options in the initialization strategy, one for MMMC (Multi-Mode, Multi-Corner) and one for a “simple” methodology. The simple methodology does not require detailed MMMC.

The Common UI initialization strategy mostly has autonomous commands, with the db being in a consistent state after each initialization step is completed. The exception to this are power intent and full MMMC timing commands. These commands are deferred until an initialization command is issued.

Types of Initialization

The following table describes the various initialization types.

Types	Description
Setup	Sets the Common UI database attributes with the specified information. However, it does not populate the database. Some basic error checking may be done, but only on the database objects.
Load	Loads the database with the specified objects. Error checking on the data being loaded can be done. The database is consistent, but is not yet ready for execution.
Init	Populates the database with the full object set and does binding between timing modes and domains. The database is fully consistent and Common UI is ready for execution.

Related Information

- [Common UI Initialization Flow and Considerations](#)
- [Flows in Common UI and the Supported Tools](#)
- [Initialization Flows - MMMC Mode and Simple](#)

Common UI Initialization Flow and Considerations

Initialization Flow

The steps in the basic flow of Common UI Initialization are explained in the table below.

Step	Tasks Performed	
1. Timing	During the Timing step, the timing libraries and constraints are specified. The Timing step is a combination of Setup and Load. The data that is loaded are the library sets pointed to by the <code>set_analysis_view</code> in the MMMC file.	
2. Physical	In this step, the physical data is loaded into the database. Both LEF and OpenAccess (OA) data are supported. Captables and QRC files are defined in the MMMC file with the <code>create_rc_corner</code> command. Therefore, loading of these is deferred to init when the full MMMC data is available. This step is optional because Synthesis and STA do not always require physical data. It is also optional for implementation to support timing regressions. The Physical Step is a Load function. All data specified in this step is loaded into the database.	
3. Design	In the Design step, the design data is in the form of a Verilog gate level netlist, or in the form of Verilog, VHDL, or System Verilog RTL. In the case of a netlist, a single command will read the netlist as well as elaborate. After this single command, the database is fully populated with the design data. In the case of RTL, there is a two-step process – reading the RTL and then elaborating. Reading the RTL is with the specific language and parameters. Elaborating is assembling the RTL into a fully-populated and consistent design view. Elaboration can also do additional RTL reads to resolve missing blocks, depending on the RTL search paths. Once elaboration is complete, the design is consistent. An optional floorplan can be loaded after the design is elaborated. Both the Innovus floorplan file format and the DEF file format are supported. If a floorplan file format is used, it must be read before the power intent is read. This ensures that the power intent power rails and definitions are used. The Design Step is always a Load function.	

4. Power	During the Power Step, the power intent is set up in the database attributes. Both CPF and IEEE 1801 are supported. In Common UI, timing intent is removed from power intent. As a result, if the CPF contains timing intent (nominal conditions or analysis views), an error will be issued unless the user specifies to ignore the timing. The Power Step is optional and only required if the design needs a power_intent file. Only db attributes are set during this step. Binding of power domains, loading of power cells, etc is not done until the Init Step.	
5. Init	The Init step is when the full database is prepared for execution. The constraints are loaded into the correct mode, the power intent is loaded and applied to the appropriate instances, and the operating conditions are applied to the power domains. Extensive error checking is performed, including checking of objects, and checking for completeness of data. Once the init step is complete, the design is fully ready for execution.	

Initialization Considerations

Following are the design initialization considerations:

- Synthesis Considerations
- Implementation Considerations
- STA Considerations

Synthesis Considerations

- Library Data
HDL elaboration requires the library data to prevent unresolved references. HDL can contain references to macros and standard cells. The current elaborator makes assumptions about port directions if a library cell is not present, and these assumptions can dramatically impact the resulting netlist. As a result, library data is needed before the design elaboration.
- Physical Data
Synthesis does not always require physical data, that is, LEF, OA or QRC files. Reading this data is optional.
- Design Manipulation
HDL elaboration can be quite complex due to various factors such as different languages and design parameters. For example, in some flows the full chip is elaborated to determine the parameters for the CPU. The design is then re-elaborated at the CPU level with the parameters. In addition, you may ungroup or rename the hierarchy before applying power intent or timing intent. Therefore, the initialization flow must support breaks for the commands after design import and before applying timing or power intent.
- Simple Mode
Both a simple setup, and an MMMC setup are supported.

Implementation Considerations

Back end is relatively simple from an initialization perspective. If timing is needed, only an MMMC setup is allowed, and design manipulation are not required. However, the design, timing, and power intent should be fully resolved before execution.

- Immediate Implementation Load
In Common UI, the implementation immediately loads some data. In Tempus legacy usage, all loading of data is deferred until `init_design`. In Common UI, specific data is loaded into the database at command invocation. This immediate load will be common across Synthesis, Implementation, and STA.
- Physical Only Flow
Common UI supports a physical only flow – using OA/LEF without the logical or timing

libraries. This is enabled by reading the physical data first without reading the timing data. When a physical-only flow is used, the timing setup and power intent setup is skipped.

Timing Analysis Considerations

- **Simple Mode**
Timing Analysis needs to support a simple non-MMMC mode. The mechanism will be identical between synthesis and Timing Analysis.
- **Physical Data**
Timing Analysis does not always require physical data, such as OA and QRC. Reading this data is optional.

Related Information

- [Stylus Common UI Initialization](#)
- [Flows in Common UI and the Supported Tools](#)
- [Initialization Flows - MMMC Mode and Simple](#)

Flows in Common UI and the Supported Tools

- [Related Information](#)

The following table shows the different flows in Common UI and the tools that they support.

Name	Description	Supported Tools		
		Genus	Innovus	Tempus
Full MMC	MMC timing + physical	√	√	√
Phys only	No <code>read_mmc</code> or <code>read_power_intent</code> <code>read_power_intent</code>		√	
MMC only	No physical	√	√	√
Simple timing	Only <code>read_libs</code> and <code>read_sdc</code>	√		√
Simple physical	Only <code>read_libs</code> , <code>read_physical</code> , <code>read_qrc</code> , and <code>read_sdc</code>	√		

Steps and Associated Commands for Common MMC Configuration

The following table shows the various commands used at different steps in different tools.

Note: The highlighted commands are optional for the particular tool.

Command	Genus	Innovus	Tempus
Timing	<code>read_mmc</code> or <code>read_libs</code>	<code>read_mmc</code>	<code>read_mmc</code>
Physical	<code>read_physical</code> or <code>read_qrc</code>	<code>read_physical</code>	<code>read_physical</code>
Design	<code>read_hdl/elaborate</code> or <code>read_netlist</code>	<code>read_netlist</code>	<code>read_netlist</code>
Floorplan	<code>read_def</code>	<code>read_floorplan</code> or <code>read_def</code>	
Power	<code>read_power_intent</code>	<code>read_power_intent</code>	<code>read_power_intent</code>

Constraints (simple mode)	<code>read_sdc</code>		<code>read_sdc</code>
Initialization	<code>init_design</code>	<code>init_design</code>	<code>init_design</code>

Related Information

- [Stylus Common UI Initialization](#)
- [Common UI Initialization Flow and Considerations](#)
- [Initialization Flows - MMC Mode and Simple](#)

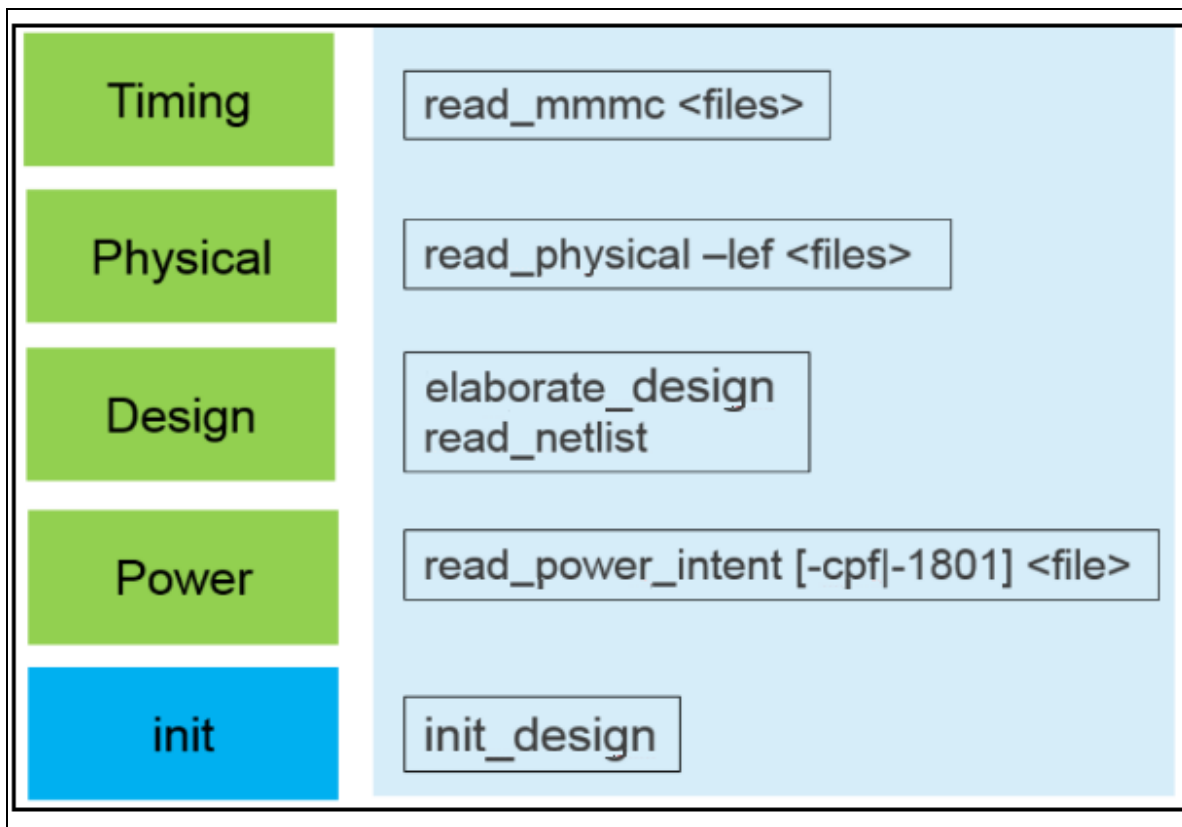
Initialization Flows - MMMC Mode and Simple

Initialization Flow - MMMC Mode

Initialization Flow for MMMC Mode includes:

- Timing
- Physical
- Design
- Power
- Init

The following image shows the initialization flow in the MMMC Mode.



Initialization Flow - MMMC Mode - Timing Step

In the Timing Step, the full MMMC objects are set up, that is, populating the basic MMMC information attributes. Synthesis needs the library information before the Design Step, therefore,

the Timing Step also loads some library information. The library data to be loaded is pointed by the `set_analysis_view` command in the MMMC file.

Initialization Flow - MMMC Mode - Physical Step

During the Physical Step, the physical libraries, OA or LEF, are loaded into the tool's database using the `read_physical` command. The database will be loaded with the specified objects, and will be available for query.

Initialization Flow - MMMC Mode - Design Step

During the Design Step, the design (netlist or RTL) is loaded into the database. There are two command sets:

`read_netlist`

- This is strictly for Verilog gate-level netlists
- After issuing this command, the design is ready for querying in the database
- This command can be used in Synthesis, Backend, and STA

`elaborate_design`

- Converts RTL to a generic netlist based on the specified script, which should contain commands for reading HDL files into memory and creating a design hierarchy consisting of a top-level design and its referenced sub-designs from Verilog or SystemVerilog modules or VHDL entity/architectures.
- The command transforms each module/architecture to be represented as a design in the form of a structural netlist.
- It also performs semantic checking, sequential register (flops/latches) inferencing, and high-level HDL optimizations. Instances of undefined modules or entities are marked as unresolved and the corresponding modules/entities reported as blackboxes.

The design step is a pure load step because in this step the database objects are populated, and can be queried and manipulated (i.e ungrouped, renamed) after this step is completed. Full error checking on the design details will be performed. This includes, but is not limited to, unresolved references, port mismatches, and empty modules.

Initialization Flow - MMMC Mode - Power Step

During the Power Step, the power intent attributes are set up and populated. This reads the specified CPF or UPF, does some basic error checking (primarily syntax checking), and sets the appropriate attributes. The design database is NOT modified – this will be done during the Init Step. In Common UI, timing intent is separated from power intent. This is not a problem with UPF as

UPF does not allow a mechanism to specify timing.

This step is a setup only step. In this step, the tool database is not loaded with the power intent objects, and no binding takes place. The reason for deferring to the Init Step is to allow further design manipulation. In this step the error checking is limited to CPF or UPF syntax checking.

After this command is issued and the user is satisfied with the results, the next step is initialization.

Initialization Flow - MMMC Mode - Init Step

The `init_design` command is an explicit command that causes the full system to be initialized. With this step, the remaining MMMC timing information is loaded, the power intent is initialized, the modes are bound to the power domains, and the design is prepared for timing. Once init is complete, the database is completely consistent and is ready for execution.

Initialization Flow - Simple

To support a simple non-MMMC initialization, there will be a simple initialization sequence consisting of simple commands. However, the underlying infrastructure and attributes are the same. Effectively these simple commands are important for the MMMC. The data set will be stored in the same MMMC attributes, and will be queried in the same way.

- Timing
- Physical
- Design
- Power
- SDC
- Init

Simple only	Timing	<code>read_lib</code>	
	Physical	<code>read_physical -lef <files></code>	
Simple only	QRC	<code>read_qrc</code>	For RC and Tempus only
	Design	<code>elaborate_design</code> <code>read_netlist</code>	
	Power	Simple power intent only – no MSV	
Simple only	<code>read_sdc</code>	<code>read_sdc</code> Creates default constraint mode	
	<code>init_design</code>	Populates corners/view Binds default mode to default corner/view	

Initialization Flow - Simple - Timing Step

In this step, libraries are loaded with the `read_libs` command. This command sets the appropriate MMMC attributes with default names.

Initialization Flow - Simple - Physical Step

In the Physical Step, LEF or OA databases are loaded via `read_physical` as done in the MMMC Physical Step. In addition, QRC files can be loaded with `read_qrc`. This will effectively do a `create_rc_corner -name default_emulate_rc_corner`, and will update the delay corner to include this default RC corner.

Initialization Flow - Simple - Design Step

This step is identical to the MMMC Design Step. The `read_def` command is also supported as an optional step. The `read_floorplan` command is not supported because Innovus does not support the simple mode.

Initialization Flow - Simple - Power Step

Power intent is not supported in the Simple mode. If `read_libs` was used to initialize the libraries, `read_power_intent` will return an error.

Initialization Flow - Simple - SDC Step

The simplified setup adds a step `read_sdc` to load in the constraints. This command accepts a list of constraints and populates the default constraint mode. No initialization is done with this step. Multiple `read_sdc` are supported. By default, each file will be appended to the default constraint mode. If you issue `-reset`, then the constraint mode will be reset, and only the new files applied.

Initialization Flow - Simple - Init Step

The Init step is the same as in the MMMC setup through `init_design`. While `read_sdc` and `init_design` could be combined into a single step, maintaining `init_design` as a separate function is good for initialization consistency. When `init_design` is used, the constraints are loaded and bound to the design elements as with the MMMC flow. Once the design has been initialized, the design is consistent and ready for timing, reporting, and implementation.

The `init_design` command should be issued only once. After the design is initialized, all subsequent updates will be done via attributes or MMMC commands.

Once the design has been initialized via `init_design`, the design is ready for execution. Execution can include incremental updates to timing and power.

Related Information

- [Stylus Common UI Initialization](#)
- [Common UI Initialization Flow and Considerations](#)
- [Flows in Common UI and the Supported Tools](#)

Database Access and Handling

- [Working with `get\_db` and `set\_db`](#)
- [Legacy vs. Stylus Common UI Database Object Commands](#)
- [Dual Ported Objects](#)
- [Significant `get\_db` Objects](#)

Working with `get_db` and `set_db`

The Stylus Common UI uses the new commands `get_db/set_db` to provide a single and common way to access/query all the database information.

These commands replace the Tempus (Legacy) commands, `dbGet`, `dbSet`, `get_ccopt_property`, and `set_ccopt_property`; and the 100s of DB object commands that have a `db` prefix such as `dbGetInstByName` and `dbSetInstPlacementStatus`. These commands also give access to all the data available from the SDC collection-based commands such as `get_cells`, `get_pins`, `get_property`, and `set_property`; although these commands are still available for SDC compatibility in the Common UI.

The `get_db` and `set_db` commands support various chaining, filtering, and matching options similar to `dbGet` and `dbSet` (in legacy) but with an easier usage model. These commands can access most object types directly from the root.

Note that `get_db` also allows string pattern matching directly on any object that has a `.name` attribute. This enables fetching the objects directly by name. For example:

```
get_db insts il/* #returns all inst objects with .name matching il/*
```

```
get_db insts -if {.name == il/*} #returns all inst objects with .name matching il/*
```

```
get_db insts .name il/* #returns all inst names (not objects) matching il/*
```

Note: `get_db` requires a space before the `.attribute` name.

Help on Database Attributes

You can also use help to find the attributes and their help description for any given `obj_type`, as

below:

```
help -obj pin      #note, "help pin:" also works, the : means only -obj help is desired,
so then -obj is not needed
Attributes:
arrival_max_fall(pin)
Returns the latest falling arrival time ...
arrival_max_rise(pin)
...
```

Pattern matching also works for help:

```
help pin: *slack*
Attributes:
slack_max(pin) Returns the worst slack across all concurrent MMMC views ...
slack_max_edge(pin) Returns the data edge of the path responsible ...
...
```

To find all the attributes with `slack` in their name, run:

```
help -attribute * *slack*
Attributes:
cts_move_clock_nodes_for_slack(clock_tree)
When set, CCOpt will consider moving nodes in the clock tree with poor ...
slack_max(pin) Returns the worst slack across all concurrent MMMC views for Setup-style
...
...
```

You can also use the <tab> key to see the attribute names of the current object when entering a `get_db` command:

```
get_db $my_obj .tab
```

Browsing the Database

You can use pattern matching for an attribute name to see multiple attribute values for many objects at the same time. This replaces the legacy `dbGet .??` usage.

So instead of this legacy usage to see all the attributes and values of every pin (`instTerm`) in

`$my_pins`, run:

```
dbGet $my_pins.??
```

You can use:

```
get_db $my_pins .*
```

Or use this to see just a few attributes of every pin:

```
get_db $my_pins .capacitance_max*
Object: pin:top/CG/BC1/Y
capacitance_max_fall: 1.0
capacitance_max_rise: 1.1
Object: pin:top/CG/BC2/A
capacitance_max_fall: 1.1
capacitance_max_rise: 1.2
...
```

If some of the values are not computed (like timing-graph data), and you want to force them to be computed anyway, run:

```
get_db $my_pins .slack_max* -computed
Object: pin:top/CG/BC1/Y
slack_max: 9227.4
slack_max_edge: rise
slack_max_fall: inf
slack_max_rise: 9227.4
...
```

Object Chaining

Object chaining enables you to link to the related objects allowed like `dbGet`.

Note: `dbGet` requires no space before the "." (`dbGet $nets.name`), but `get_db` requires at least one space before the "."

For example:

- Pins or ports that drive a specific pin:
`get_db pin:top/rst_reg/D .net.drivers`
- Pins or ports that are loads of a specific pin:
`get_db pin:top/rst_reg/D .net.loads`

Examples

- The following command finds all the insts that start with `i1/i2/` and end with `_buf`:
`get_db insts i1/i2/*_buf`
`inst:top/i1/i2/test_buf inst:top/i1/i2/i3/test_buf ...`
- The following command counts the number of repeater cells:
`llength [get_db insts -if { .base_cell.is_buffer || .base_cell.is_inverter }]`
- The following `get_db` command uses `-foreach` to count the number of `base_cell` used in the `netlist` inside the `cell_count` Tcl array.
`get_db insts .base_cell -foreach {incr cell_count ($obj(.name))}`

```
array get cell_count      #write out the array with the counts
BUF1 20 AND2 30 ...
```

- The following command defines a user-defined attribute, and then adds it on a net based on its fanout:

```
define_attribute high_fanout \
-category test \
-data_type bool \
-obj_type net
set_db [get_db nets -if {.num_loads > 20}] .high_fanout true
get_db net:clk .high_fanout
true
```

- The following command count all the pins below the hinst, or all pins at one level of the hierarchy

```
llength [get_db hinst:tdsp_core/ALU_32_INST .insts.pins]
llength [get_db hinst:tdsp_core/ALU_32_INST .local_insts.pins]
```

- The following command filters out the buffer cells, and counts them in buf_count:

```
get_db insts -if {.base_cell.is_buffer==true} \
-foreach {incr buf_count($obj(.base_cell.name))}
array get buf_count
BUF1X 20 BUF2X 30 ...
```

- The following command returns all the placed sequential cells:

```
get_db insts -if {.base_cell.is_sequential==true && .place_status==placed}
```

Legacy vs. Stylus Common UI Database Object Commands

Legacy database object commands are object-specific commands, and `dbGet` access uses `-p1` to get the object pointer while searching a name. `dbGet` also chains through the top or head objects and sometimes through other objects, whereas the `get_db/set_db` commands can access most object types directly from the root.

Note that `get_db` also allows string pattern matching directly on any object that has a `.name` attribute. This makes it easy to get objects directly by name. For example:

```
get_db insts il/* #return all inst objects with .name matching il/*
```

```
get_db insts -if {.name == i1/*} #return all inst objects with .name matching i1/*
get_db insts .name i1/* #return all inst names (not objects) matching i1/*
```

Some examples of usage of the legacy and stylus common UI database object commands are:

Legacy	Stylus Common UI
dbFindInstsByName i1/*	get_db insts i1/*
dbGet -p1 top.insts.name i1/*	get_db insts i1/*
dbGet -p1 top.fplan.netGroups my_grp	get_db net_groups my_grp
dbGet [dbGet -p1 head.layers.name metall].width	get_db [get_db layers metall] .width
get_property [get_pins i1/p1] arrival_max_fall	get_db [get_db pins i1/p1] .arrival_max_fall
get_ccopt_property target_max_trans - clock_tree clk1 -early	get_db [get_db clock_trees clk1] .cts_target_max_transition_time_early

Note: `get_db` requires a space before the `.attribute` name, while `dbGet` does not allow a space.

References

For more information, see:

- [dbGet](#)
- [dbSet](#)
- [get_db](#)
- [set_db](#)

Dual Ported Objects

The Stylus Common UI uses the Tcl Dual Ported Objects (DPO) concept. A DPO is a Tcl_Obj in the Tcl C++ programming interface. The object pointer is kept as a C++ pointer inside Tcl unless Tcl forces it to be converted to its "dual" string form. In normal usage it never gets converted to a string which is more efficient, but if you do a `puts $var` or return the value to the shell to echo to the `xterm`, it is converted to its string form.

- The string form is normally the name of the object preceded by the `obj_type`, so a layer object string form might look like `layer:metal1`.
- A design object name has the current design name included, so an instance named `i1/i2` in design `top`, looks like `inst:top/i1/i2`.
- An object with no name like a wire, just shows the pointer hex-value like `wire:0x22222222`.

`get_db/set_db` commands allow both a DPO list and a collection from SDC commands like `get_pins` as input, but only returns objects in a DPO Tcl list.

For example, if your design is called `top`, then the output is something like this:

```
set i1_insts [get_db insts i1/*]
inst:top/i1/i2 inst:top/i1/i3 ...
```

You can also use the DPO name directly for input. So the last three queries in the table above could be done more easily this way:

Using Returned DPO Value	Using DPO Name Directly
<code>get_db [get_db layers metal1] .width</code>	<code>get_db layer:metal1 .width</code>
<code>get_db [get_db pins i1/p1] .arrival_max_fall</code>	<code>get_db pin:top/i1/p1 .arrival_max_fall</code>
<code>get_db [get_db clock_trees clk1] .cts_target_max_transition_time_early</code>	<code>get_db clock_tree:top/clk1 .cts_target_max_transition_time_early</code>

References

For more information, see:

- [get_db](#)

- [set_db](#)
- [Stylus Common UI Database Object Information](#) manual

Significant get_db Objects

Most of the Common UI object names are similar to the legacy object names used by `dbGet` except that they use all lower-case characters, with `_` for word separators, and the Common UI names use fewer abbreviations (like `pd` is now `power_domain`, and `bndry` is `boundary`).

Objects or names that were changed substantially are listed below:

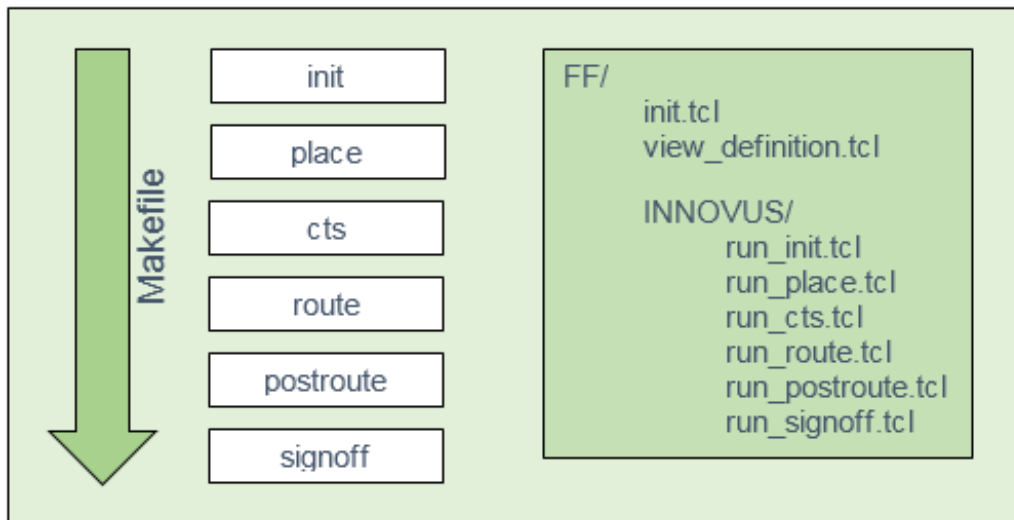
dbGet Object Name	get_db Object Name	Comments
<code>topCell</code>	<code>design</code>	A top-level design (the top-level Verilog module). Note that <code>topCell fplan</code> and <code>hinst</code> attributes are directly in the design object (such as <code>rows</code> or <code>hinsts</code>); there is no longer a <code>topCell fplan</code> object, or the <code>topCell hinst</code> object.
<code>head</code>	<code><none></code>	Not used. The head attributes (like <code>layers</code>) are directly available from the root.
<code>fplan</code>	<code><none></code>	Not used. The <code>topCell.fplan</code> attributes (such as <code>rows</code> , <code>blockages</code> , and <code>bndrys</code>) are directly in the design object.
<code>libCell</code>	<code>base_cell</code>	A library <code>base_cell</code> (like <code>AND2</code>) created from the LEF and Liberty <code>.lib</code> definitions. This is the library data that does not vary with different timing corners, so pin names, LEF attributes, and <code>.lib</code> attributes (like <code>is_flop</code>) are here. Most <code>base_cells</code> have multiple <code>lib_cells</code> (<code>.lib</code> data) attached for each timing corner, unless it is a physical-only <code>base_cell</code> like a filler cell that has no <code>lib_cell</code>. So every <code>inst</code> has a <code>base_cell</code>, but every <code>base_cell</code> does not have a <code>lib_cell</code>.

<code>term(libCell)</code>	<code>base_pin</code>	A logical pin on a library <code>base_cell</code> .
<code>term(topCell)</code>	<code>port</code>	A logical port on the design. The design and <code>base_cell</code> usage was separated into two different objects, because they have many attributes that only apply to one of the two cases.
<code><none></code>	<code>lib_cell</code>	A Liberty (<code>.lib</code>) defined library cell. There can be many <code>lib_cells</code> with different timing data for one <code>base_cell</code> .
<code><none></code>	<code>lib_pin</code>	A Liberty (<code>.lib</code>) defined library cell pin. There can be many <code>lib_pins</code> with different timing data for one <code>base_pin</code> .
<code>instTerm</code>	<code>pin</code>	A logical pin on an <code>inst</code> .
<code>hinstTerm</code>	<code>hpin</code>	A logical pin on the outside of an <code>hinst</code> (it connects to <code>hnets</code> outside the <code>hinst</code>).
<code>hTerm</code>	<code>hport</code>	A logical port on the inside of an <code>hinst</code> (it connects to <code>hnets</code> inside the <code>hinst</code>).
<code><none></code>	<code>analysis_view</code> <code>clock</code> <code>library</code> <code>library_set</code> <code>opcond</code> <code>rc_corner</code> <code>timing_condition</code> <code>arc</code> <code>lib_arc</code> <code>timing_point</code> <code>timing_path</code> <code>path_group</code>	Timing objects accessed with SDC commands, such as <code>get_pins</code> and <code>get_property</code> , are available.

<none>	clock_tree clock_tree_source_group clock_spine flexible_htree skew_group	cchopt objects accessed with legacy get_cchopt_property are available.
prop	<direct access>	A property value is now directly accessed using the property name. So instead of: <pre>dbGet [dbGet -p1 \$inst.props.name my_prop_name].value</pre> you can use: <pre>get_db \$inst .my_prop_name</pre>
viaRuleGenerate	via_def_rule	A via definition rule (like a LEF VIARULE GENERATE statement).
via	via_def	A via definition.
viaInst	via	An instance of a via.
sViaInst	special_via	An instance of a special via.
<none>	obj_type attribute	The DB schema is available with these objects. This replaces the legacy dbSchema command. You can do this instead: <pre>get_db obj_types #shows all obj_types</pre> <pre>get_db attributes *slack* #shows all attributes that have slack in their name</pre> <pre>help -attribute *slack* #shows formatted help output for all attributes with slack in their name</pre>

Foundation Flows

The foundation flow is a code generation engine that takes a Tcl variable array (vars) that defines both the library/design data as well as tool options or flow control and generates a set of scripts to execute a baseline flow. Each flow step is a single Tcl script that gets executed via a Makefile. An example of a simple example flow (default baseline flow) and the scripts that execute the flow is shown below:



The flow can be broken down into two areas: design initialization and design execution. Design initialization takes place as part of `run_init.tcl` and is supported by the `FF/init.tcl` and `FF/view_definition.tcl` files, which define the design and library data required to construct the initial database.

In addition, the flow can be customized through the use of tags and plug-ins that allow you to augment or override the flow commands within the sequence of flow steps. It means that tag and plug-in files are included as part of the flow step script.

The Foundation Flow has been replaced with a more integrated approach to flow construction and execution that works across the entire flow seamless in the Stylus Common UI.

Related Information

- [Flowkit](#)
- [flowtool](#)

Flowkit

Flowkit enables the previous step scripts (standalone Tcl files) become flow objects, which are stored in the database. In the example below, the flow block is defined as a flow of sub-flows, which map to the same basic flow sequence as the foundation flow:

```
create_flow -name block {flow:fplan flow:prects flow:cts flow:route flow:postroute}
```

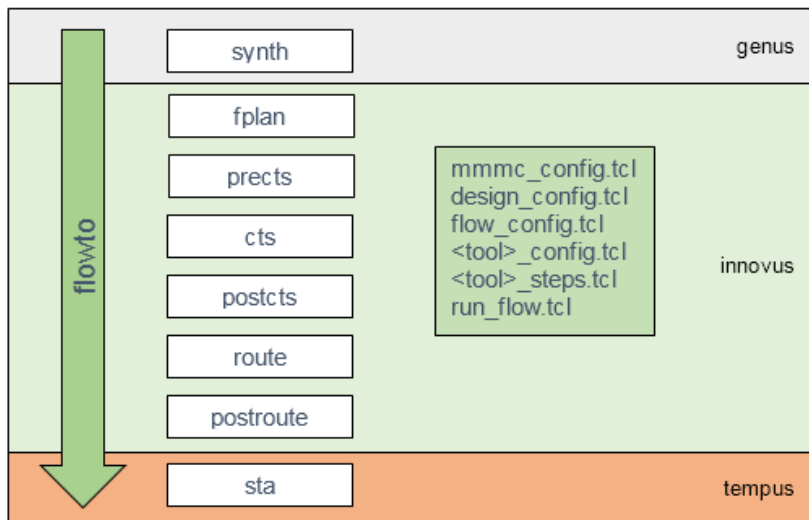
Also, there is `design_config.tcl`, which contains the design initialization commands and the `mmmc_config.tcl`, which is the flowkit equivalent of the `view_definition.tcl` files.

The `tool_steps.tcl` file contains the `create_flow_step` commands for the steps associated with a particular tool (Genus, Innovus, or Tempus). For a multi-tool flow (as depicted below), the block flow would be pre-pended with `flow:synth` (genus) and post-pended with `flow:sta` (tempus).

Since the flow definitions for the (sub)flows specify the tool, flowkit knows how to launch them. Example:

```
create_flow -name prects -tool tempus {flow_start run_place_opt flow_finish}
```

The Flowkit Stylus Flow



Each of the sub-flow consists of the flow steps. The flow customization is done by creating the specified flow steps and attaching them before or after one of these flow steps.

For example, the previous `pre_cts` plug-in in the foundation flow would be handled in the following way. Consider a plug-in file (`plug/pre_cts.tcl`) that contains the following:

```
set_interactive_constraint_modes [all_constraint_modes -active]
set_clock_uncertainty -setup 0.050 [get_clocks {CLK}]
set_clock_uncertainty -hold 0.003 [get_clocks {CLK}]
reset_propagated_clock [all_clocks]
reset_propagated_clock [get_ports [get_property [all_clocks] sources]]
```

The stylus flowkit `cts` flow is defined as the following sequence of steps:

```
create_flow -name cts -tool tempus \
{flow_start add_clock_spec build_clock_tree add_tieoffs flow_finish}
```

You can create a flow step that contains the plug-in commands (if necessary, convert the legacy plug-in commands to Common UI commands and use `convert_legacy_to_common_ui`).

```
create_flow_step -name pre_cts {
set_interactive_constraint_modes [all_constraint_modes -active]
set_clock_uncertainty -setup 0.050 [get_clocks {CLK}]
set_clock_uncertainty -hold 0.003 [get_clocks {CLK}]
reset_propagated_clock [all_clocks]
reset_propagated_clock [get_ports [get_property [all_clocks] sources]]
}
```

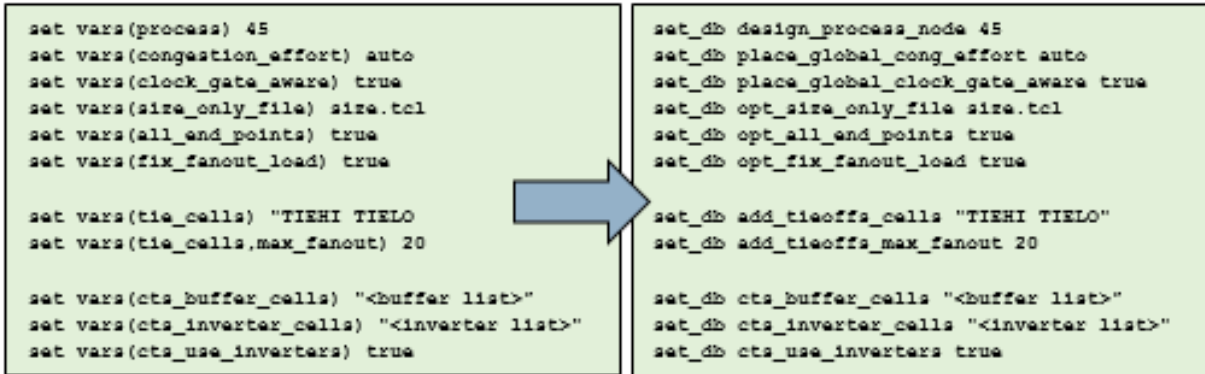
This flow step would then be added to the flow before the `add_clock_spec` flow step:

```
edit_flow -append flow_step:pre_cts -before flow_step:add_clock_spec
```

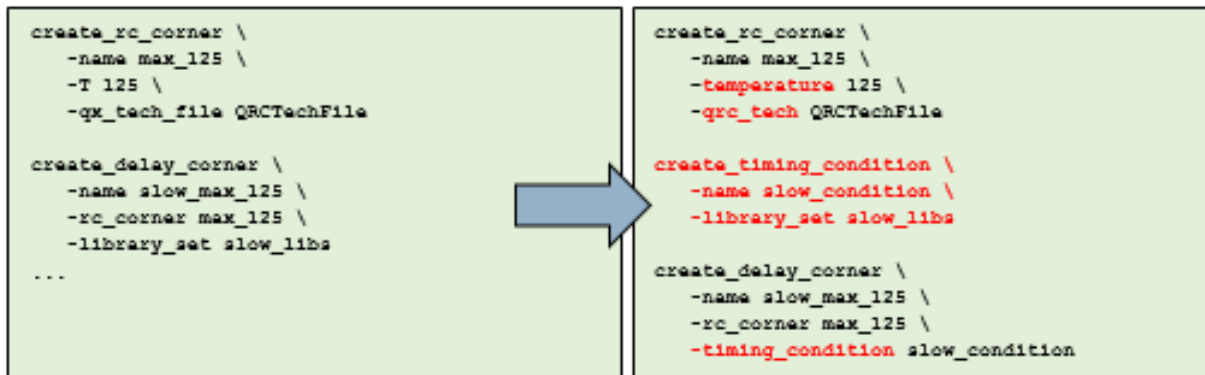
The flow execution is managed by a new executable called `flowtool`, as show below, so make is no longer necessary.

The foundation flow `setup.tcl` contained variable definitions that the code generator would use to create the command sequence and determine the flow step sequence (for example, `place_opt_design` and `fix_hold`). With a flowkit, `write_flow_template` allows you to enable features that control the step content and the sequence of the flow templates that are generated.

With respect to the variable definition in the `setup.tcl` that is used to map commands or command options, the large majority of these now map directly to Common UI attributes, which makes the need for a layer of abstraction unnecessary:



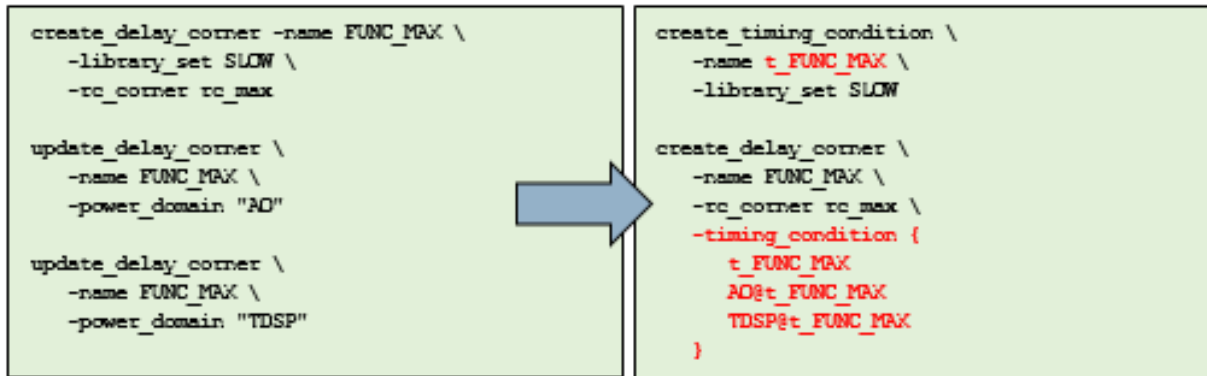
Additionally, the rest of the vars were used to define the timing information for the MMMC view definition file generation (for example, `library_sets`, `rc_corners`, `delay_corners`, `constraint_modes`, and `analysis_views`). The Common UI (and hence the flowkit) requires additional migration of the view definition file. In particular, the new MMMC syntax has a new object called the timing condition. Here is an example of converting a legacy MMMC file to CUI syntax:



The timing condition object defines the operating condition and library set:

Parameter	Description
<code>-help</code>	Displays the command usage.
<code>-library_sets <i>string</i></code>	Specifies the list of libsets to associate to this timing condition.
<code>-name <i>tc_name</i></code>	Specifies unique name for the timing condition.
<code>-opcond <i>string</i></code>	Specifies single name of an operating condition.
<code>-opcond_library <i>string</i></code>	Specifies name of library to search for the named opcond.

In addition, the `timing_condition` argument to `create_delay_corner` supports the new syntax for power domain binding:



The command `update_delay_corner` is no longer necessary because the power domains are bound directly to a timing condition using the `@` syntax.

flowtool

`flowtool` is an external command in Unix for running flows to work interactively within a Tcl session and is used for:

- Hierarchical flows in which multiple independent block flows run in parallel.
- Each step of a flow in its own session to improve productivity (in the absence of session reset).
- Flows that cover required tools (in particular, RTL-GDSII flows).

In addition to providing another interface to the existing `flowkit` functionality, the `flowtool` enables multi-session, multi-tool, and branching flows.

Make versus flowtool

The command behavior of `make` varies significantly across platforms (BSD and GNU), and the capabilities of GNU in particular varies significantly across versions. This can be mitigated by writing to a lowest common denominator, but still creates a testing issue.

`make` targets are the files that are regenerated based on other files. This model works well for compiling C, but not for the complex flows. Currently, the foundation flow gets around this problem by creating dummy files corresponding to steps run. This works, but can lead to unexpected behavior if you do not know the location. Another strategy is to use the databases saved at the end of steps instead of the dummy files. This is less clear in the cases where steps do not create a database (notably, partition).

A significant problem with the `make` approach is that details of the flow, most importantly the flow order, are embedded in the Makefile. When these details change, the Makefile must be regenerated which contradicts the desire to perform `codegen` only once. The complexity of using `make` in a dynamic flow environment can be avoided by developing a tool which is more tailored to running flows. `flowtool` is specialized for running flows across multiple tools and multiple sessions and overcomes the limitations described previously.

`flowtool` has the following advantages over `make`:

- `flowtool` is entirely self-contained and independent of the flow. The `make` approach requires a flow-specific Makefile to be generated for each project.
- You can modify the command line syntax based on the specific task, unlike being restricted to

the make syntax.

- This approach reduces or eliminates dependencies on the deployment environment. For example, you can implement it as a Tcl script, and allows shipping the Tcl interpreter, or bundling the two together to make a standalone executable.
- Dependency tracking and task scheduling do not need to be based on files (although state would probably still be stored in the file system).
- Interface to machine distribution (LSF, SSH, etc.)

Benefits of flowtool

The benefits of using `flowtool` include:

- No code is generated; `flowtool` is a static script or executable.
- Flow information is only stored in the database, so cannot become inconsistent.
- The next steps can be determined. This is important for the partitioning flow, as it allows the partitioning step itself to determine the number and inputs of its successors at runtime, based on the partitioning it has performed.

Timing Reports

The Stylus Common UI interface provides consistent UI and output for timing reports that improve the ability to analyze and debug the timing information. The timing reports are designed to accomplish the following goals:

- Easier and faster identification of issues.
- Customized reports for analyzing specific problems.
- Additional content to identify certain challenges (such as, power domains, congestion, etc).
- Consistent report output as all the timing report functions are accessible in the same way.
- Better use of hardware resources.

Related Information

- [Reporting Commands](#)
- [Report Output Formatting](#)
- [Supported Reporting Attributes](#)

Reporting Commands

The reporting commands, mainly `report_timing` provide enhanced syntax for better clarity.

The commands support text and CSV output formats that can be redirected to a file using the standard redirect characters (that is, `>` and `>>`). The default output is text to `stdout`.

For more information on the supported reporting commands used for generating timing reports, see the [Timing Analysis Commands and Attributes](#) chapter.

Related Information

- [Report Output Formatting](#)
- [Supported Reporting Attributes](#)

Report Output Formatting

The timing reports follow a standard formatting structure. The report output contains the following sections:

- Path Summary Section
- Path Description Section

The report sections are explained below.

Path Summary Section

The path summary section provides a quick snapshot of each reported path that allows you to identify various problems easily and quickly.

Path Description Section

The path description section shows the cell-by-cell delay, slew, arc, exception information, and informational flag legends.

By default, the timing reports show the 'path summary' and 'path description' section. However, the default can be changed by using the `report_timing -path_type` parameter.

Sample Report

A sample path description section is shown below.

```
Path 1: MET (3595 ps) Setup Check with Pin FF_O2_reg[0]/CK->D
Group:      ck1
Startpoint: (R) FF_I1_reg[0]/CK
Clock:      (R) ck1
Endpoint:   (F) FF_O2_reg[0]/D
Clock:      (R) ck1

          Capture      Launch
Clock Edge:+    9000          0
Src Latency:+    0          0
Net Latency:+    0 (I)    0 (I)
Arrival:=       9000          0

Setup:-         318
Uncertainty:-   125
Required Time:= 8557
Launch Clock:-  0
Data Path:-     4962
Slack:=         3595

#-----
# Timing Point          Arc      Edge      Cell          Load      Trans      Delay      Arrival
#                      (fF)      (ps)      (ps)      (ps)
#-----
FF_I1_reg[0]/CK        <<< -      R          -              -          150         -           0
FF_I1_reg[0]/Q         CK->Q      R          DFFHQX4        45.4        201        334        334
mul_26_28/Fn0755D13176/Y A->Y      F          INVX2          15.7        103         68        402
mul_26_28/Fn0755D13172/Y A->Y      R          INVX4          53.0        223        130        532
mul_26_28/n0617DTn0525/Y B->Y      F          NAND2X1        10.9        169        108        640
mul_26_28/n0634Dtn0548/Y A1N->Y    F          AOI2BB2X4      19.3        164        232        872
mul_26_28/Fn0612D/Y    A->Y      R          INVX2          18.5        183        119        992
. . .
mul_26_43/n0014Dtn0331/Y A->Y      F          NAND2X2        17.3        147         94       4168
mul_26_43/Fn0013D16832/Y A->Y      R          INVX1           5.6        145         97       4265
mul_26_43/n0078Dtn0331/Y A1N->Y    R          OAI2BB1X2     18.9        201        203       4468
mul_26_43/n0064Dtn0331/Y A->Y      R          XOR2X4         10.6         86        225       4693
mul_26_43/n0025Dtn0331/S CI->S      F          ADDFHX4         5.2         82        269       4962
FF_O2_reg[0]/D        <<< -      F          DFFHQX2         -           -           0       4962
#-----
```

Endpoint Reports

In addition to detailed reports, endpoint reports can also be generated (using the `report_timing -path_type endpoint` option). These reports allow you to include additional endpoint related information that can be useful in debugging or identifying certain issues. For example, an endpoint report can show the number of buffers and inverters on a path.

A sample report is shown below:

```
> report_timing -path_type endpoint

#-----
# ID      Slack      Endpoint Group View
# (ns)
#-----
1      -0.034255    out      clk      view3
2      -0.034255    out      clk      view5
3      -0.034255    out      clk      view4
4      0.389435     f0/D     clk      view3
5      0.389435     f0/D     clk      view4
6      0.389435     f0/D     clk      view5
7      0.398548     f1/D     clk      view5
8      0.398548     f1/D     clk      view4
9      0.398548     f1/D     clk      view3
#-----
```

The endpoint reports and detailed reports are mutually exclusive.

Use of Informational Flags

Informational tags are used providing additional details. The following informational tags are supported:

Flags	Description
(i)	Net is ideal
(P)	Instance is preserved / don't_touch
(p)	Instance is preserved / don't_touch but may be resized

The output format uses the flag keys on the timing report and adds a description at the end of the report for reference. A sample output is as shown below:

```
#-----
# Timing Point  Delay (ns) Latch  Window  Arrival  Arc  Flags  Adjustment
# (ns)
#-----
f0/CP          -          -      1.000000          CP      -      -
f0/Q           0.254100  -      1.254100          CP->Q   -      -
b32/Z          0.598446  -      1.852546          I->Z    -      -
...
b4/Z           0.086940  -      2.502181          I->Z    -      -
f1/D           0.000000  -      2.502181          D       <<<    -
#-----

(P) : Instance is preserved
(p) : Instance is preserved but may be resized
(i) : Net is ideal.
```

(b) : Timing paths are broken.

Retime Analysis Report Header Flags

AOCV(default)	Indicates that the selected retime analysis includes recalculation of path depths and AOCV derates. The AOCV retime is enabled with the <code>-retime aocv</code> or <code>aocv_path_slew_propagation</code> parameter. The AOCV stage counting mode (set using <code>timing_aocv_analysis_mode</code>) is indicated in the parenthesis. The stage counting modes are given below: • <code>launch_capture(default)</code> (Default) • <code>combine_launch_capture</code> (Combine) • <code>separate_clock_data</code> (Separate) • <code>clock_only</code> (Clock)
Corr	Indicates that the current SOCV accuracy mode considers correlation of variation between stages. Starting with SOCV accuracy mode <code>medium</code> (specified with the <code>-socv_accuracy_mode</code> parameter), it is assumed that some portion of the variation is not completely uncorrelated random variation, so the software tracks a correlated and uncorrelated component for each of the applicable moments.
Data Clock	Indicates the portion of the path that has been retimed. By default, only the data path is retimed. If the <code>timing_disable_retime_clock_path_slew_propagation</code> global variable is set to <code>false</code>, then the clock path will also be retimed, and "Clock" will be added to the retime status.
EWM	Indicates that equivalent waveform model is used (<code>-equivalent_waveform_model</code> parameter).
GBA COLL	Shows a collection of paths for which retime analysis is not performed. This can be controlled with the When the <code>report_timing -path_type exhaustive -collection</code> command. By default, the <code>report_timing -collection</code> parameter does not show GBA and PBA slack values separately.
GP	Indicates that graph-based PBA mode settings are enabled.
GS	Indicates that GBA skewness mode settings are enabled.
Path-Slew	Indicates the selected retime analysis is set to use path correct slew rates to reduce the pessimism from slew merging. This can be controlled with the <code>report_timing -retime</code> parameter. The path slews are recalculated in <code>path_slew_propagation</code> and <code>aocv_path_slew_propagation</code> modes.
PBA COLL	Shows a collection of paths for which retime analysis is performed while creating the collection. This can be controlled with the When the <code>report_timing -path_type exhaustive -collection</code> command. By default, the <code>report_timing -collection</code> parameter does not show GBA and PBA slack values separately.
PG	Indicates that group-based mode reporting is enabled when N paths are reported for every group.
RC	Indicates that RC variation is enabled with the <code>timing_socv_rc_variation_mode</code> setting, and an SOCV format RC variation side file library is available.

SI	Indicates that SI analysis is enabled during the retime analysis. The path-based analysis will recompute SI effects using path correct victim timing windows.
SOCV(Ultra)	Indicates that SOCV analysis is enabled and has been incorporated into the retiming. The SOCV analysis mode (specified with the <code>-socv_accuracy_mode</code> parameter) is shown in parenthesis. The accuracy modes are mapped as follows: • low <not shown> • medium Medium • high High • ultra Ultra
Spatial (CS)	Indicates that spatial derating side files have been loaded and an explicit spatial derate factor will be calculated. This is not shown for AOCV 2D derate tables, where the spatial and depth derates are combined. The GBA spatial derate mode (specified with the <code>timing_spatial_derate_distance_mode</code> setting) is shown in parenthesis with the possible values given below: • chip_size (CS) • bounding_box (BB) The GBA spatial derate mode does not affect the path-based analysis bounding box calculation.
WP	Indicates that waveform propagation is enabled (using the <code>set_delay_cal_mode</code> equivalent <code>ent_waveform_model</code> propagation setting). The path-based retime will also use this delay calculation accuracy setting when recomputing delays and transitions for the path.

Examples of GBA/PBA Collection Reports

- When the `report_timing -path_type exhaustive -collection` command is specified, the following command generates a GBA collection report as shown below:

```
tempus > report_timing $tim_coll ; set tim_coll [report_timing -path_type exhaustive -collection]
```

```
**WARN: (TCLCMD-1248): Since no '-max_slack' has been specified with exhaustive '-retime_mode', tool will
assume it's default value in this case which is '0'
```

```
**WARN: (TCLCMD-1251): The PBA exhaustive mode '-nworst' limit has been exhausted for some of the
endpoints. Refer to verbose log file of current session for more details.
```

```
INFO: Path Based Analysis (PBA) performed on total '3' paths
```

```
INFO: Max depth 3 reached for end point 'htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/E' during Path Based Analysis
```

```
Path 1: VIOLATED Clock Gating Setup Check with Pin htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/CK
```

```
Endpoint: htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/E (v) checked with leading edge of 'clk'
```

```
Beginpoint: htb/mc_vl_reg_2/Q (v) triggered by leading edge of 'clk'
```

```
Common point: SPINE_CELL_BBF1/Y
```

```
Path Groups: {clock_gating_default}
```

```
Analysis View: sg_05v_15c_sigmax
```

```
Retime Analysis {GBA COLL}
```

- When the `report_timing -path_type exhaustive -collection` command is specified, the following command shows a PBA collection report:

```
tempus > report_timing $tim_coll ; set tim_coll [report_timing -path_type exhaustive -collection]
```

```
**WARN: (TCLCMD-1248): Since no '-max_slack' has been specified with exhaustive '-retime_mode', tool will
assume it's default value in this case which is '0'
```

```
INFO: Path Based Analysis (PBA) performed on total '530' paths
```

```
INFO: Max depth 131 reached for end point 'htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/E' during Path Based Analysis
```

```
Path 1: VIOLATED Clock Gating Setup Check with Pin htb/RC_CG_REBALANCE_HIER_INST2989/RC_IC_IN_1/CK
```

```
Endpoint: htb/RC_CG_REBALANCE_HIER_INST2989/RC_IC_IN_1/E (v) checked with leading edge of 'clk'
```

```
Beginpoint: htb/mc_vl_reg_2/Q (^) triggered by leading edge of 'clk'
```

```
Common point: SPINE_CELL_BBF1/Y
```

```
Closest CPPR Driver Pin: { Launch: SPINE_CELL_BBF1/Y Capture: SPINE_CELL_BBF1/Y }
```

```
Path Groups: {clock_gating_default}
```

```
Analysis View: sg_05v_15c_sigmax
```

```
Retime Analysis {PBA COLL}
```

VT Skew Reports

The following informational tags are supported for VT skew reports:

Flags	Description	Command Settings
VT-Skew	Shows VT skew details.	<code>set timing_enable_vtskew_derate_mode true</code>
PBA	Shows that the current path has undergone VT skew analysis. However, all the paths might not be considered VT skew candidates - even if VT skew analysis mode is enabled.	<code>set_vt_skew_config -unique_vt_mode {none slow fast nominal}</code>
PU	Indicates unique VT skew for PBA. Displayed when unique VT mode is enabled.	<code>set_vt_skew_config -unique_vt_mode {slow fast nominal}</code>
GU	Indicates unique VT skew for GBA. Displayed when unique VT mode is enabled.	

Sample Report

- The following report shows report header flags for retiming analysis:

```

Path 1: MET Hold Check with Pin TDSP_CORE_INST/EXECUTE_INST/top_reg[11]/CP
Endpoint:   TDSP_CORE_INST/EXECUTE_INST/top_reg[11]/D           (v) checked
with leading edge of 'm_clk'
Beginpoint: TDSP_CORE_INST/DATA_BUS_MACH_INST/data_out_reg[11]/Q (v) triggered
by leading edge of 'm_clk'
Path Groups: {m_clk}
Analysis View: view2

Retiming Analysis {Data Path-Slew SOCV(High)-Corr SI Vt-Skew (PBA GU PU)}
Retiming Vt-Skew Analysis {VT_group_1@fast VT_group_2@fast VT_group_3@fast }
Retiming Vt-Skew Stats { Dominant_VT_group: NA Launch_Clock_VTGroups: 2 Capture_
Clock_VTGroups: 1 Total_Clock_VTGroups: 2}

Other End Arrival Time      0.569769 (+3.00S) | 0.531640 | 0.012710
+ Hold                      -0.035400 (+3.00S) | -0.035400 | 0.000000
+ Phase Shift               0.000000
- CPPR Adjustment          0.051262 (+3.00S) | 0.000100 | 0.017054
= Required Time             0.462027 (+3.00S) | 0.496140 | 0.011371 (-)
  Arrival Time              0.769317 (-3.00S) | 0.803430 | 0.011371
  Slack Time                0.307290 (-3.00S) | 0.307290 | 0.000000
= Slack Time(original)      0.356572 (-3.00S) | 0.356572 | 0.000000
  Clock Rise Edge           0.000000
  = Beginpoint Arrival Time  0.000000 (-3.00S) | 0.000000 | 0.000000

```

Supported Fields

By default, timing reports show default fields. Any additional fields can be added (or removed) by using the `report_timing -fields` parameter.

The supported fields and their default status is shown in the table below:

Field	Description	Detailed Report	Endpoint Report
-------	-------------	-----------------	-----------------

adjustment	Reports generated clock adjustment values. When reporting the detailed path for a generated clock, there are circumstances where the arrival time needs to be adjusted when a <code>create_generated_clock</code> assertion is encountered along the path. The adjustment argument can be used to provide a clearer indication of why a "jump" in the arrival time is seen in the path report. There are two primary reasons for generated clock-related adjustment to occur. • If the <code>timing_enable_generated_clock_edge_based_source_latency</code> attribute is set to <code>false</code>, it is possible that the edge-to-edge requirement specified by the generated clock cannot be satisfied by an existing logic path. This can happen when the source clock to generated clock polarity of the constraint is non-inverting, but the clock path itself constrains an inverting path. In this case, it would be normal to see a half-cycle of adjustment as the software compensates for the path versus constraint mismatch. If the above mentioned attribute is set to true, the software handles the path versus constraint mismatch by zeroing the source latency of the generated clock, and forcing alignment of the source and generated clock. As a result, no further adjustment is required, or reported. • In cases where <code>create_generated_clock -edge_shift</code> is specified, the adjustment column will show the amount of shift specified.		
abs_delay_err	Reports the absolute delay difference of Tempus from Spice.		
abs_transition_err	Reports the absolute slew difference of Tempus from Spice.		
annotation	Reports RC parasitic annotations for net arcs, and delay annotations for either gate or net arcs. The following annotations are reported: • SDF: SDF back-annotation. The annotation status reflects the gate arc, not the interconnect arc. • SPEF: This entry appears if parasitic extraction has been done or annotated using SPEF. • LumpedRC: Lumped RC annotation • DlyAssert: Assertion set with <code>set_annotated_delay</code> command • WLM: Wire load model back-annotation • <none>: All others		
aocv_adj_stages	Shows up "Aocv Adjust Stages" column. This column reports the AOCV adjustment stages that indicate the total effective stage count.		
aocv_derate	Reports the graph-based AOCV derating factor used for cells/nets in AOCV mode. In retime mode, this column will report graph-based derating for cells/nets before the AOCV branch point and path-based derating.		
aocv_weight	Reports AOCV weight values for the path elements.		

arc	Reports the arc as described by the from pin, from pin edge, to pin, and to pin edge. For example, the arc from the rising edge of pin A to the falling edge of pin Z is reported as A -> Z.	Y	
arrival	Reports the arrival time on the pin.	Y	
arrival_mean	Reports mean of the arrival time on a pin.		
arrival_sigma	Reports standard deviation of the arrival time on a pin.		
cell	Reports the cell name of the given pin's instance.	Y	
delay	Reports the arc delay. If the stolen slack at the arc output pin (output of transparent latches) is not zero, the stolen column is also displayed along with this column.	Y	
delay_mean	Reports the mean of each arc's delay.		
delay_sigma	Reports the standard deviation of each arc's delay. Note: In the absence of correlated and uncorrelated components of variation, the individual delay sigma values will not be summed up.		
direction	Reports the pin direction (IN, OUT).		
edge	Reports the edge of a pin (^=R, v=F).	Y	
fanin	Reports the number of source nodes of a net connected to a timing pin.		
fanout	Reports the number of sinks of a net connected to a timing pin.	Y	
flags	Indicates the size_only, dont_use, dont_touch, and ideal attributes. When the 'flags' option is specified, a capacitance group entry (if it exists) will be displayed under the "Flags" column. This value indicates the input capacitance group being used for delay calculations – C1C2 (2-piece model) or C1CN (n-piece model).	Y	
hpin	Reports the hierarchical name for a given pin. Note: The hpin, pin, and timing_point will print the same values in the report output.		
incr_delay	Reports the incremental delay of a pertaining arc. The software reads the incremental delay from an incremental SDF file using the <code>read_sdf</code> command.		
instance	Reports the hierarchical name of a given pin's instance.	Y	
instance_location	Reports the location (x,y) of an instance. If a design is not placed, this column will be blank. This option displays the <code>instance</code> column automatically.		
latch_window	Reports the available window for latch to sample the data.		
load	Reports the total capacitance load on a given pin.		
module	Reports the module name of the corresponding block hierarchy. This hierarchical module name will be reported only when the <code>-hpin</code> parameter of the <code>report_timing</code> command is specified.		
net	Reports the hierarchical name of a net connected to a given pin.		

pct_delay_err	Reports the percent delay difference of Tempus from Spice.		
pct_slew_err	Reports the percent slew difference of Tempus from Spice.		
phase	Reports the phase name of a pin.		
phys_info	Shows minimum/maximum assigned layer, and non-default rule name information.		
pin	Reports the reference name for the given pin.		
pin_load	Reports the pin load data. The pin load refers to the sum of pin capacitance of all the pins connected to the current net.		
pin_location	Reports the location (x,y) of a pin. If a design is not placed, this column will be blank.		
power_domain	Reports power domain details of the specified path instance.	Y	
required	Reports the required time on a pin.		
retime_delay	Reports the recalculated arc delay based on the path being reanalyzed with the specified retime method.		
retime_delay_mean	Reports the mean of the recalculated arc delay on the path being reanalyzed with the specified retime method.		
retime_mis_delta	Reports MIS delta delay in PBA mode for MIS-aware timing arcs in the path.		
retime_mis_derate	Reports MIS derate values (in PBA mode), when derate-based advanced multi-input switching (AMIS-Derate) analysis is enabled.		
retime_delay_sigma	Reports the standard deviation of each parameter of an arc's retimed delay.		
retime_slew	Reports the recalculated slew based on the path being reanalyzed with the specified retime method.		
retime_slew_mean	Reports the mean of each parameter of an arc sink pin's retimed slew.		
retime_slew_sigma	Reports the standard deviation of each parameter of an arc sink pin's retimed slew.		
stage_count	Displays the aggregate stage count of a specific instance in graph-based analysis (GBA) mode, and the actual stage count in path-based analysis (PBA) mode.		
transition	Reports the propagated slew at the given pin.		
transition_mean	Reports mean of each parameter of an arc sink pin's slew.		
transition_sigma	Reports the standard deviation of each parameter of an arc sink pin's slew.		
socv_derate	Reports the SOCV or the spatial derating factor applied on the cells or nets in SOCV analysis mode.		
spice_arrival	Reports the Spice arrival time at the timing point.		

spice_delay	Reports the Spice delay of a timing arc.		
spice_slew	Reports the Spice slew of a timing point.		
stolen	Reports the slack stolen (or the time given to the previous stage) at the given pin. If the slack is not zero and the <code>delay</code> column is specified, then this column is displayed by default (but only visible on the output of transparent latches).		
timing_point	Reports the hierarchical names for the given points.	Y	
total_derate	Reports the total derating factor applied on the corresponding cell or nets that reflect the combined impact of all the derates (OCV/AOCV/spatial) applied on each stage of the timing path. Note: By default, the software will take into account the negative delays and keep the SOCV derate column in sync with the user-derate column. The software multiplies the negative delay by (2.0 - derating factor) to make it more pessimistic. However, reporting will always show derates based on the positive delay. The timing report output will show similar values for the SOCV Derate and Total Derate columns, but there will be no changes in the slack values.	Y	
user_derate	Reports the derating scale factors set with the <code>set_timing_derate</code> command. Note: The timing system cannot provide an accurate derating factor for the lumped gate and wire delay when the <code>set_timing_derate</code> factors for <code>-cell_delay</code> and <code>-net_delay</code> are not equivalent. In this case, an asterisk (*) is reported in the SSI Derate column.		
voltage	Returns the voltage that is used during delay calculations.		
vt_skew_derate	Returns voltage threshold (VT) skew analysis details.		
when_cond	Reports the "when" condition of the arc specified in the library.		
wire_load	Reports the wire load.		
wlmodel	Reports the kind of parasitic used by a pin or net, such as SPEF. If the wire load model is derived from a Liberty file, the information is reported in the following format: <pre>Wireload model (.lib model_name library_name)</pre>		

Related Information

- [Reporting Commands](#)
- [Supported Reporting Attributes](#)

Supported Reporting Attributes

You can use the `timing_report_fields` attribute to control the default output of timing reports - that is, the report fields and options can be changed.

For more information on attributes, see [Timing Category Attributes](#).

The `get_db` command can be used to obtain information associated with the default and added fields. For example, the following command shows the added fields:

```
> get_db timing_report_fields
```

Output:

```
timing_point flags arc edge cell fanout transition delay arrival
```

Related Information

- [Reporting Commands](#)
- [Report Output Formatting](#)

Uniform Logging

Command logging plays a vital role in the debug process. The Stylus Common UI provides improved and uniform logging across products by logging all commands in the log file, irrespective of whether they are issued interactively or through startup files and scripts.

In Common UI, logging is verbose by default for all commands, except those originating through user procs. The following table shows how logging works for commands issued using various methods:

Command origin	Logging type	Log Output Example
Interactive typing	Always logged directly in log file	@tempus 1> report_timing
Sourcing files	Verbose logging, by default To turn off verbose logging for all source file requests, set the <code>source_verbose</code> attribute to <code>false</code> To turn off verbose logging for a specific source file request, use <code>source -quiet file.tcl</code>	<pre>@tempus 1> source scripts/setup.tcl #@ Begin verbose source /proj/scripts/setup.tcl @file 1: set_db design_process_node 16 ... Applying the recommended capacitance filtering threshold values for 16nm process node: total_c_th=0, relative_c_th=1 and coupling_c_th=0.1. @file 2: set_db opt_fix_hold_verbose true @file 3: set_db route_design_skip_analog true #@ End verbose source /proj/scripts/setup.tcl 1 true</pre>
Running flow_steps	Verbose logging, by default To turn off verbose logging, set the <code>flow_verbose</code> attribute to <code>false</code>	<pre>@tempus 1> create_flow_step -name timing { report_timing } @tempus 2> create_flow -name rpt {flow_step:timing} @tempus 3> run_flow -flow rpt ... #@ Begin verbose flow_step rpt.timing @flow 1: report_timing ##### # Generated by: Cadence Tempus 20.10-b015_1 # OS: Linux x86_64(Host ID noi-leenap1) # Generated on: Tue Feb 11 16:48:49 2020 # Design: carve # Command: report_timing ##### ... #@ End verbose flow_step rpt.timing</pre>

Calling procs	Non-verbose logging, by default To enable verbose logging, use the <code>set_proc_verbose</code> command	Default log output example <pre>@tempus 1> proc hi {} {puts Hello} @tempus 2> hi Hello</pre> Output with <code>set_proc_verbose</code> <pre>@tempus 1> define_proc foo {} { place_opt_design; report_timing } @tempus 2> set_proc_verbose foo @tempus 3> foo #@ Begin verbose proc foo @proc 1: place_opt_design ## Begin place_opt_design ## End place_opt_design @proc 2: report_timing ##### # Generated by: Cadence Tempus 17.10-d119_1 # OS: Linux x86_64(Host ID sj-jasond) # Generated on: Wed Mar 15 10:37:56 2017 # Design: dtmf_recvr_core # Command: report_timing ##### #@ End verbose proc foo</pre>
Issued through GUI	Similar to interactive typing	On selecting <i>Timing -> Report Timing</i> from the GUI menu, log file will show: <pre>@tempus 1> report_timing</pre>
Product startup files	Verbose logging, by default To turn off verbose logging, set the shell envar <code>CDS_STYLUS_SOURCE_VERBOSE</code> to 0, which will set <code>source_verbose</code> to false at startup	If you add the following to the <code>./cadence/tempus/tempus.tcl</code> startup file: <pre>puts "inside [info filename]"</pre> the log file, by default, shows: <pre>#@ Loading startup files ... @tempus 1> source ./cadence/tempus/tempus.tcl #@ Begin verbose source ./cadence/tempus/tempus.tcl @file 1 : puts "inside [info filename]" inside ./cadence/tempus/tempus.tcl #@ End verbose source ./cadence/tempus/tempus.tcl</pre>

Product startup options (-files and -execute)	Verbose logging, by default To turn off verbose logging, set the shell envar <code>CDS_STYLUS_SOURCE_VERBOSE</code> to 0, which will set <code>source_verbose</code> to false at startup	If the file <code>run.tcl</code> contains the following: <pre>proc test {} {puts "This is a test"} test</pre> and the product is started with: <pre>tempus -stylus -files run.tcl</pre> the log file, by default, shows: <pre>#@ Loading startup files ... @tempus 1> source .cadence/tempus/tempus.tcl #@ Begin verbose source .cadence/tempus/tempus.tcl @file 1 : puts "inside [info filename]" inside ./cadence/tempus/tempus.tcl #@ End verbose source .cadence/tempus/tempus.tcl #@ Processing -files option @tempus 2> source run.tcl #@ Begin verbose source /proj/scripts/run.tcl @file 1: proc test {} {puts "This is a test"} @file 2: test This is a test #@ End verbose source /proj/scripts/run.tcl</pre>
---	--	--

Uniform Startup

In Stylus Common UI, the bootstrap files are loaded in a consistent order across all tools - Genus, Innovus, Joules, Modus, Tempus, and Voltus. The startup sequence is listed in the table below:

#	Startup File	Example
1	<code>~/.cadence/<tool>/{gui.pref.tcl, gui.color.tcl, workspaces}</code> in the home directory if the GUI is enabled.	<code>~/.cadence/innovus/{gui.pref.tcl, gui.color.tcl, workspaces}</code>
2	<code>.cadence/<tool>/{gui.pref.tcl, gui.color.tcl, workspaces/}</code> in the current directory if the GUI is enabled.	<code>.cadence/innovus/{gui.pref.tcl, gui.color.tcl, workspaces/}</code>
3	<code><install_dir>/etc/<tool>/<tool>.tcl</code> in the installation directory. Note: For Tempus and Voltus, <code>etc/<tool></code> is <code>etc/ssv</code> because they share the same installation. However, both these tools use separate Tcl file names - <code>tempus.tcl</code> and <code>voltus.tcl</code> , respectively.	<code><install_dir>/etc/innovus/innovus.tcl</code>
4	<code>~/.cadence/<tool>/<tool>.tcl</code> in the home directory.	<code>~/.cadence/innovus/innovus.tcl</code>
5	<code>./.cadence/<tool>/<tool>.tcl</code> in the current directory.	<code>./.cadence/innovus/innovus.tcl</code>
6	<code>./<tool>_startup.tcl</code> in the current directory.	<code>./innovus_startup.tcl</code>

Note: The `-stylus` usage sets the `source_verbose` root attribute to `true` at startup. The GUI preference init files (1 and 2 above) ignore this setting, but the non-GUI preference initialization files

above (3, 4, 5, and 6) will use verbose logging by default. If the shell `envvar CDS_STYLUS_SOURCE_VERBOSE` exists and has a boolean value, then the `source_verbose` attribute will be set to the `envvar` value at startup. So you can turn off verbose logging at startup by using the following command:

```
setenv CDS_STYLUS_SOURCE_VERBOSE 0
```

Metrics

- [Stylus Common UI Metrics](#)
- [Metric Commands](#)
- [Metric Object Definition](#)
- [Snapshots](#)
- [Snapshots Hierarchy and Inheritance](#)
- [Streaming Metrics](#)

Stylus Common UI Metrics

Unified Metrics (UM) is a system integrated with some Cadence tools that automatically delivers data about the design and run to you. UM provides functionality to:

- Stream data from the reports and algorithms.
- Collect and organize the stream data into a structured form.
- Display the data in a Simple Unified Metrics file or the Advanced Unified Metrics server.

Currently, the primary digital flow tools, Genus, Innovus, Voltus, and Tempus are UM-enabled as an integral part of the Stylus Common UI feature across these tools.

Stylus Common UI Metric Infrastructure

- In Common UI, metrics are captured and stored in a file format common to various tools, such as Innovus and RC.
- There are common sets of metrics that are captured consistently in different tools as well as tool-specific metrics.
- The metrics file/database can be used to generate reports/graphs in various formats and can be compared against other runs.
- Metrics can be queried using the `get_metric` command and are written to a `<design>.metrics` file in the Innovus database directory as part of `write_db`.

Using the Common UI metric infrastructure, you can define and capture metric snapshots, using the `create_snapshot` command, at specific points in the flow such that metric extraction is not required, and/or timing/power analysis will be calculated at the point the snapshot is saved.

- Timing/power metrics are reported based on the last calculated value. You need to ensure that the metrics are updated prior to saving the metric snapshot. All commands that save metrics call `create_snapshot` to save metrics in a consistent manner.
- Because this data will get saved as part of the database (during `write_db`), it can be used across multiple runs for building design project summary pages.

Metrics Overview

A metric is the core representative of a piece of data. A metric has a name and a value. The value should have an SI unit specified wherever possible. The TCL command for setting a metric is `set_metric`.

- The following command would set the value of the worst negative setup slack (all path groups) to 0.100 ns.

```
set_metric -name timing.setup.wns -value {0.100ns}
```

- The following commands would set the value of the total area of the design to 100 um.

```
set_metric -name design.area -value {100um}
```

You can get a metric using the `get_metric` command.

- The following command would get the value of the worst negative setup slack (all path groups), which is set to 0.100 ns.

```
% get_metric -name timing.setup.wns  
0.100 ns
```

Metric Names

Metric names are standardized across the tools so that the same name will contain the same data. This means that each tool will populate a metric with a value that represents the same semantic value. This value can be compared across the runs. The names are look hierarchical with the use of the ".". Use of "." allows grouping common metrics through their names. For example, consider the following metric names and how they could be represented as a hierarchy.

	flow
flow.cputime 6.96	-- cputime
flow.cputime.total 6.96	-- total
flow.cputime.user 6.33	`-- user
flow.cputime.user.total 6.33	`-- total
flow.machine vlsj-ben1	-- machine
flow.machine.load 0.81	`-- load
flow.memory 373.004	-- memory
flow.memory.resident 26.4375	`-- resident
flow.realtime 63	-- realtime
flow.realtime.total 63	`-- total
flow.tool.name Innovus Implementation System	`-- name
flow.tool.name.short innovus	`-- short

Metric Values

Metric values can be of many types but are most often a number. Wherever possible, the unit should be defined so that the metric display can always show the values in the same units. This will prevent different tools from producing numbers in different units.

Related Information

- [Metric Commands](#)
- [Metric Object Definition](#)
- [Snapshots](#)
- [Snapshots Hierarchy and Inheritance](#)
- [Streaming Metrics](#)

Metric Commands

The following table shows some metric commands in Common UI:

Command	Description
<code>define_metric</code>	Allows the user to define new metrics
<code>create_snapshot</code>	Creates a snapshot of all the defined metrics
<code>get_metric</code>	Gets the current value of a metric or metrics
<code>read_metric</code>	Loads a previously saved metric file
<code>write_metric</code>	Writes out metrics in various file formats
<code>report_metric</code>	Generates reports for the specified metrics

Related Information

- [Stylus Common UI Metrics](#)
- [Metric Object Definition](#)
- [Snapshots](#)
- [Snapshots Hierarchy and Inheritance](#)
- [Streaming Metrics](#)

Metric Object Definition

The following table lists the metric object definition in Common UI:

Attribute	Type	Description
<code>name</code>	String	Name of the metric
<code>value</code>	String	Value of the metric

visibility	Enum	Default, user, hidden, internal
default_unit	String	Default unit for the metric (assumed for value)
unit	String	Unit to use for reporting
threshold	Float	Threshold for numeric metrics
threshold_function	Enum	How to interpret the threshold (greater_than, less_than)
precision	Int	Number of decimal places
tcl	TCL	List of TCL commands or proc to generate (return) the metric that is of the format { <value> <unit> }

Related Information

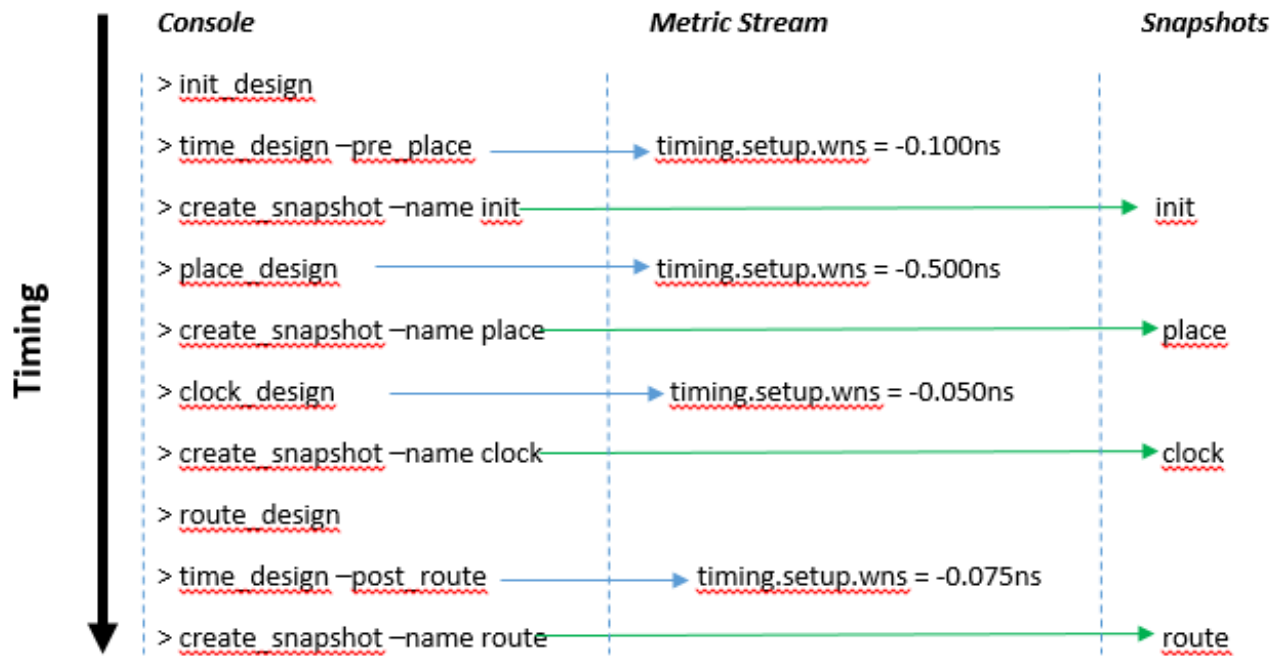
- [Stylus Common UI Metrics](#)
- [Metric Commands](#)
- [Snapshots](#)
- [Snapshots Hierarchy and Inheritance](#)
- [Streaming Metrics](#)

Snapshots

The metric stream is collected and structured with the use of snapshots. A snapshot records all the current values of the metrics and stores them in the database. You can create a snapshot by using the `create_snapshot` command.

```
create_snapshot -name my_first_snapshot
```

You can use snapshots to check what happened during the flow:



In the above example, four snapshots, init, place, clock, and route were created. You can check this in the Simple Unified Metrics html file and the Advanced Unified Metrics server in a table. For example:

Snapshot	Setup WNS
clock	-0.050 ns
init	-0.100 ns
place	-0.500 ns
route	-0.075 ns

Related Information

- [Stylus Common UI Metrics](#)
- [Metric Commands](#)
- [Metric Object Definition](#)
- [Snapshots Hierarchy and Inheritance](#)
- [Streaming Metrics](#)

Snapshots Hierarchy and Inheritance

Snapshots Hierarchy

You can use snapshots to produce a hierarchy, which allows you to check the data of a specific step. You can do this by managing the *snapshot stack* to control the current depth of the snapshots being made.

For example the following code would produce the following tables:

```
> set_metric -name timing.setup.wns -value {-0.100 ns}
> create_snapshot -name init
> push_snapshot_stack
> set_metric -name timing.setup.wns -value {-0.700 ns}
> create_snapshot -name place_start_up
> set_metric -name timing.setup.wns -value {-0.550 ns}
> create_snapshot -name place_work
> set_metric -name timing.setup.wns -value {-0.500 ns}
> create_snapshot -name place_finish
> pop_snapshot_stack
> create_snapshot -name place
> set_metric -name timing.setup.wns -value {-0.050 ns}
> create_snapshot -name clock
> set_metric -name timing.setup.wns -value {-0.075 ns}
> create_snapshot -name route
```

The above code creates the following snapshot hierarchy:

```
run
|--- init
|--- place
| |--- place_start_up
| |--- place_work
```

```
| `--- place_finish  
|--- clock  
`--- route
```

In the html file, a table is produced with a link for the "place" snapshot. When you click this link, it will show the lower hierarchy steps that were included through the use of the stack.

Snapshot	Metric		Snapshot	Metric
init	-0.100 ns		place_start_up	-0.700 ns
place	-0.500 ns	click link →	place_work	-0.550 ns
clock	-0.050 ns		place_finish	-0.500 ns
route	-0.075 ns			

Note: While using the stack it is important to remember that the system is working with a stream of data. Therefore, when you push, a lower level of hierarchy is created, and when you pop the stack, the generated child snapshots are included in the **next** snapshot to be created.

Snapshots Inheritance

When a hierarchy is produced, the parent snapshot automatically *inherits* metric values from the last child snapshot. For example, consider the following:

```
> set_metric -name m -value 1  
> create_snapshot -name before_parent  
> push_snapshot_stack  
> set_metric -name m -value 2  
> create_snapshot -name child_A  
> set_metric -name m -value 3  
  
> create_snapshot -name child_B  
> pop_snapshot_stack  
> create_snapshot -name parent  
> set_metric -name m -value 4  
> create_snapshot -name after_parent
```

The above code creates the following snapshot hierarchy:

```
run
|--- before_parent (m=1)
|--- parent (m=3 *inherited)
| |--- child_A (m=2)
| `--- child_B (m=3)
`--- after_parent (m=4)
```

In this example, the snapshot "parent" will inherit the value for M that was captured by snapshot "child_B", that is, '3'.

Values are only inherited if that metric is not explicitly set during the parent. For example, consider the slight modification of the above example:

```
> set_metric -name m -value 1
> create_snapshot -name before_parent
> push_snapshot_stack
> set_metric -name m -value 2
> create_snapshot -name child_A
> set_metric -name m -value 3

> create_snapshot -name child_B
> pop_snapshot_stack
> set_metric -name m -value PARENT_VALUE; ##### <--- added line
> create_snapshot -name parent
> set_metric -name m -value 4
> create_snapshot -name after_parent
```

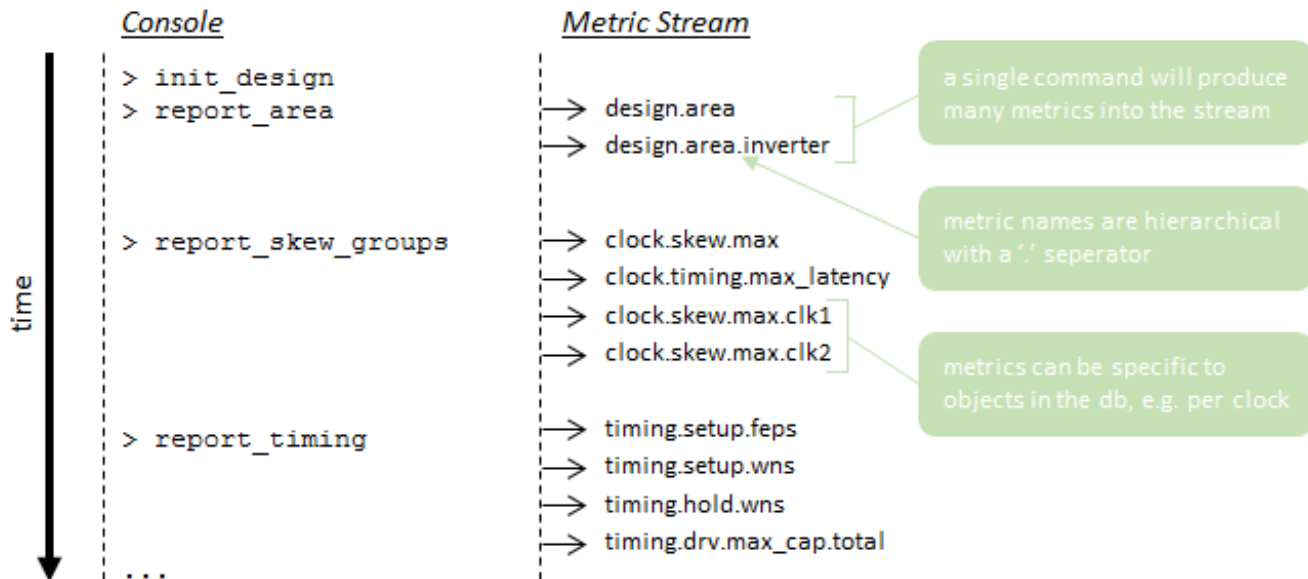
This would result in the value of 'PARENT_VALUE' being captured for the 'm' metric in the 'parent' snapshot.

Related Information

- [Stylus Common UI Metrics](#)
- [Metric Commands](#)
- [Metric Object Definition](#)
- [Snapshots](#)
- [Streaming Metrics](#)

Streaming Metrics

The UM-enabled tools will automatically produce data as it is available. For example, the diagram below shows the execution (in this case within Innovus) of some of the TCL commands and the example metrics that are produced into the metric stream. If the same metric name is produced, only the latest value is remembered by the tool, that is, the metrics will naturally overwrite each other.



Related Information

- [Stylus Common UI Metrics](#)
- [Metric Commands](#)
- [Metric Object Definition](#)
- [Snapshots](#)
- [Snapshots Hierarchy and Inheritance](#)